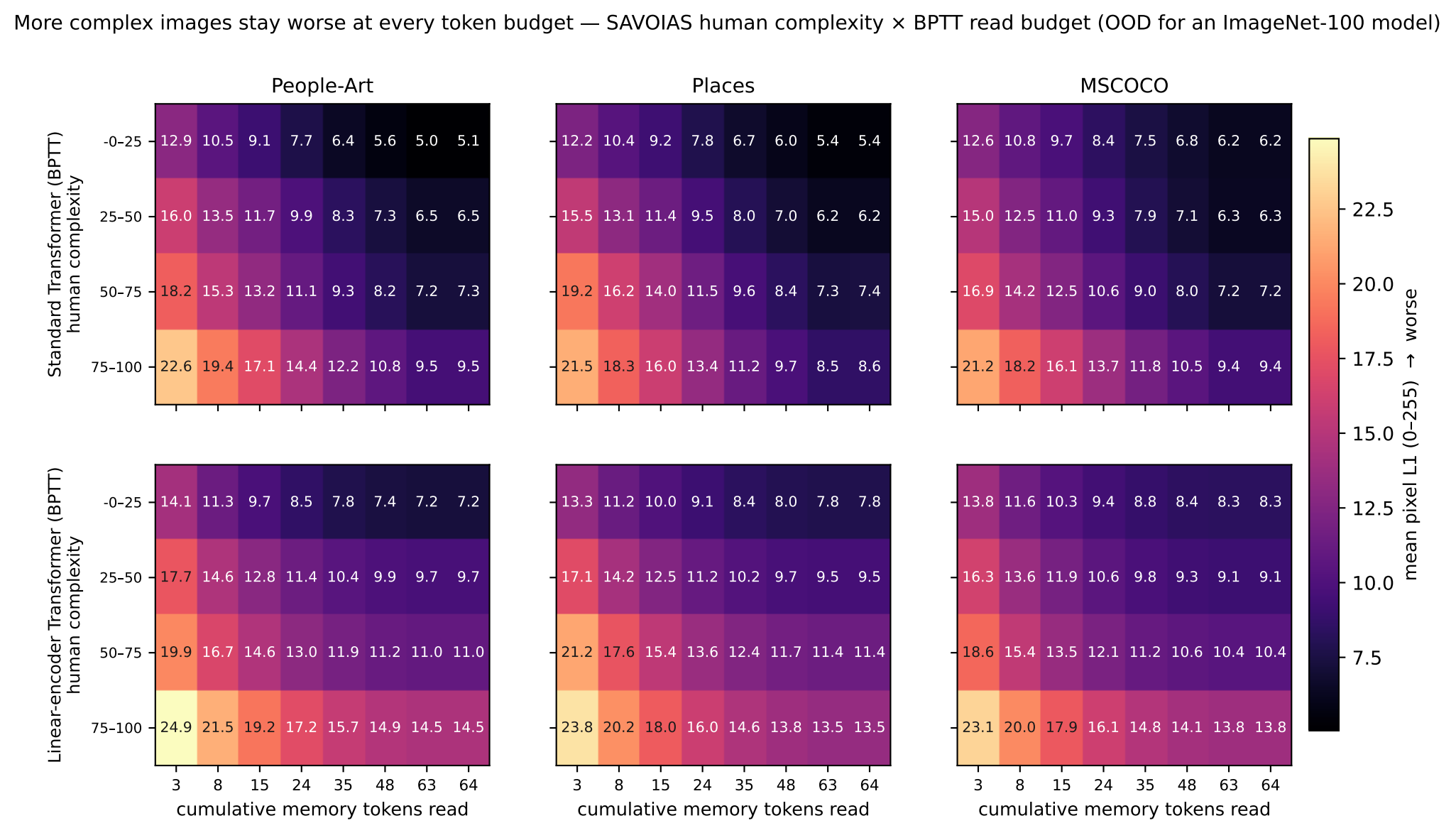
More complex images stay worse at every token budget — SAVOIAS human complexity × BPTT read budget (OOD for an ImageNet-100 model)
People-Art
Places
MSCOCO
Standard Transformer (BPTT)
human complexity
Linear-encoder Transformer (BPTT)
human complexity
-0–25
25–50
50–75
75–100
12.9 10.5 9.1 7.7 6.4 5.6 5.0 5.1
16.0 13.5 11.7 9.9 8.3 7.3 6.5 6.5
18.2 15.3 13.2 11.1 9.3 8.2 7.2 7.3
22.6 19.4 17.1 14.4 12.2 10.8 9.5 9.5
12.2 10.4 9.2 7.8 6.7 6.0 5.4 5.4
15.5 13.1 11.4 9.5 8.0 7.0 6.2 6.2
19.2 16.2 14.0 11.5 9.6 8.4 7.3 7.4
21.5 18.3 16.0 13.4 11.2 9.7 8.5 8.6
12.6 10.8 9.7 8.4 7.5 6.8 6.2 6.2
15.0 12.5 11.0 9.3 7.9 7.1 6.3 6.3
16.9 14.2 12.5 10.6 9.0 8.0 7.2 7.2
21.2 18.2 16.1 13.7 11.8 10.5 9.4 9.4
14.1 11.3 9.7 8.5 7.8 7.4 7.2 7.2
17.7 14.6 12.8 11.4 10.4 9.9 9.7 9.7
19.9 16.7 14.6 13.0 11.9 11.2 11.0 11.0
24.9 21.5 19.2 17.2 15.7 14.9 14.5 14.5
13.3 11.2 10.0 9.1 8.4 8.0 7.8 7.8
17.1 14.2 12.5 11.2 10.2 9.7 9.5 9.5
21.2 17.6 15.4 13.6 12.4 11.7 11.4 11.4
23.8 20.2 18.0 16.0 14.6 13.8 13.5 13.5
13.8 11.6 10.3 9.4 8.8 8.4 8.3 8.3
16.3 13.6 11.9 10.6 9.8 9.3 9.1 9.1
18.6 15.4 13.5 12.1 11.2 10.6 10.4 10.4
23.1 20.0 17.9 16.1 14.8 14.1 13.8 13.8
3 8 15 24 35 48 63 64
cumulative memory tokens read
mean pixel L1 (0–255) → worse
22.5
20.0
17.5
15.0
12.5
10.0
7.5